

Bitcoin Market Sentiment vs Trader Performance Analysis

Author

Pranav Mahadev Raikar

Objective

The objective of this project is to analyze the relationship between Bitcoin market sentiment and trader performance using the Fear & Greed Index and Hyperliquid historical trading data. The analysis aims to uncover how different market sentiment conditions influence trading activity, profitability, and trader behavior.

Datasets Used

1. Bitcoin Fear & Greed Index Dataset

Columns:

- timestamp
- value
- classification
- date

Sentiment Categories:

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

2. Hyperliquid Historical Trader Dataset

Columns:

- Account
- Coin
- Execution Price
- Size Tokens
- Size USD
- Side
- Timestamp IST

- Closed PnL
- Fee
- Trade ID
- Transaction Details

Total Records: 211,224 trades

Methodology

Data Preprocessing

1. Loaded both datasets using Pandas.
2. Checked for missing values and data quality issues.
3. Converted timestamp columns into datetime format.
4. Extracted date information from trade timestamps.
5. Merged the trading dataset with the Fear & Greed dataset using the Date field.

Exploratory Data Analysis

The following analyses were performed:

- Trade activity by sentiment
- Average profit by sentiment
- Total profit by sentiment
- Win-rate analysis
- Position size analysis
- Profit distribution visualization

Results and Findings

Trading Activity by Sentiment

Sentiment	Number of Trades
Fear	61,837
Greed	50,303
Extreme Greed	39,992
Neutral	37,686
Extreme Fear	21,400

Observation

Fear periods recorded the highest trading activity, while Extreme Fear recorded the lowest. This suggests traders remain active during moderate uncertainty but become more cautious during severe market pessimism.

Average Profit per Trade

Sentiment	Average Closed PnL (\$)
Extreme Greed	67.89
Fear	54.29
Greed	42.74
Extreme Fear	34.54
Neutral	34.31

Observation

Extreme Greed generated the highest average profit per trade, indicating strong momentum-driven opportunities during highly optimistic market conditions.

Total Profit Generated

Sentiment	Total Closed PnL (\$)
Fear	3.36 Million
Extreme Greed	2.72 Million
Greed	2.15 Million
Neutral	1.29 Million
Extreme Fear	0.74 Million

Observation

Fear periods generated the highest overall profits despite market uncertainty. This suggests many traders successfully capitalize on market pullbacks and volatility.

Key Insights

Insight 1

Fear periods produced the highest cumulative profit of approximately \$3.36 million, making them the most profitable market condition overall.

Insight 2

Extreme Greed delivered the highest average profit per trade at \$67.89, highlighting the effectiveness of momentum-based trading during bullish sentiment.

Insight 3

Trading activity peaked during Fear periods with over 61,000 trades, indicating strong participation during uncertain market conditions.

Insight 4

Extreme Fear experienced the lowest participation and profitability, suggesting increased trader caution and reduced market confidence.

Insight 5

Market sentiment significantly influences trader behavior, trading volume, and profitability.

Conclusion

This study demonstrates a clear relationship between Bitcoin market sentiment and trading performance. Fear periods generated the highest total profits and trading activity, while Extreme Greed offered the highest average profit per trade.

The findings indicate that traders respond differently under varying emotional market conditions. Incorporating sentiment indicators into trading strategies can provide valuable context for decision-making and risk management.

Recommendations

1. Monitor Fear & Greed Index values as a supplementary trading signal.
2. Consider momentum-based strategies during Extreme Greed periods.
3. Explore contrarian opportunities during Fear periods where cumulative profitability is highest.
4. Apply stronger risk management during Extreme Fear periods.
5. Combine sentiment analysis with technical indicators for improved trading decisions.

Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib

- Seaborn
- Jupyter Notebook

Project Outcome

The project successfully identified meaningful relationships between market sentiment and trader profitability, providing actionable insights that can support smarter and more informed trading strategies.